5.6 The Fermat Point

WORKSHEET

The question

STEP III 2000, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q3

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Where are A, B and C? What are their arguments? What angles do OA, OB and OC make with each other?

STAGE 2 Equilateral triangles as rotations

Ask yourself Triangle CAN is equilateral, lettered clockwise. If you stand at C, how do you get from A to N? Express $N - C$ in terms of $A - C$.

Pause and try Find L and M in the same way, and simplify using the identities from Stage 1.

STAGE 3 Three lines through the origin

Ask yourself Compare $L = \alpha(p + q)$ with $C = -r\alpha$. What do you notice? Do the same for M and A, and for N and B.

Try this next

1. **Check the lettering.** Show that if the equilateral triangles were lettered anticlockwise (built inwards), the three lines would generally not pass through O.

2. **Napoleon's theorem.** Show that the centre of the equilateral triangle ABL is $\frac{1}{3}(A + B + L)$. Using the expressions above, show that the three centres form an equilateral triangle.

3. **The minimum.** With $p = 2$, $q = 1.4$, $r = 1.1$, compute $PA + PB + PC$ for P at each vertex and at the centroid $\frac{1}{3}(A + B + C)$, and check that all exceed 4.5.

4. **An equilateral test.** Show that z_1, z_2, z_3 form an equilateral triangle (in either orientation) if and only if $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$.

Further reading

- **Richard Courant & Herbert Robbins, *What Is Mathematics?*** (Oxford University Press). Chapter 7 discusses the Fermat point, Steiner trees and soap films.
- **Liang-shin Hahn, *Complex Numbers and Geometry*** (Mathematical Association of America). Many geometric theorems proved with complex numbers, at exactly this

level.

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 1 includes Napoleon's theorem and other geometric results.

The full guided solution, and what the question leads to at university, is in Section 5.6:
https://beyondstep.keithmcnulty.org/ch05/fermat-point.html